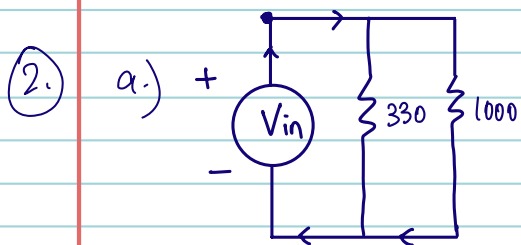


EDES 301: ASSIGNMENT #1

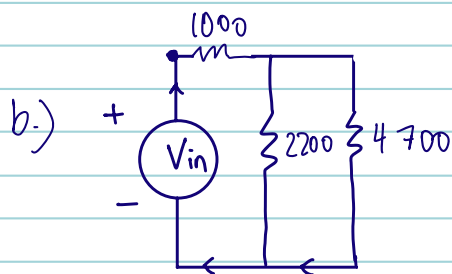
- 1)
- a.) $22\Omega \rightarrow$ Red, Red, Black, Gold
 - b.) $330\Omega \rightarrow$ Orange, Orange, Brown, Gold
 - c.) $1k\Omega \rightarrow$ Brown, Black, Red, Gold
 - d.) $2.2k\Omega \rightarrow$ Red, Red, Red, Gold
 - e.) $4.7k\Omega \rightarrow$ Yellow, Violet, Red, Gold
 - f.) $10k\Omega \rightarrow$ Brown, Black, Orange, Gold
 - g.) $100k\Omega \rightarrow$ Brown, Black, Yellow, Gold



$$250\Omega \rightarrow R_1 \parallel R_2 = 248.12\Omega$$

$$R_1 = 330\Omega$$

$$R_2 = 1k\Omega$$

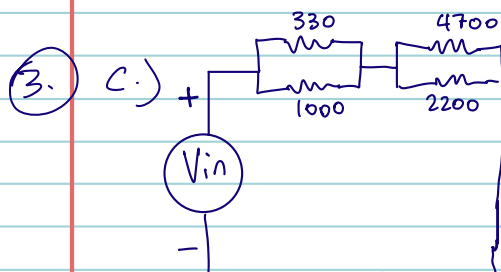


$$2500\Omega = R_1 + (R_2 \parallel R_3) = 2498.55\Omega$$

$$R_1 = 1k\Omega$$

$$R_2 = 4.7k\Omega$$

$$R_3 = 2.2k\Omega$$



$$1750\Omega \Rightarrow (R_1 \parallel R_2) + (R_3 \parallel R_4)$$

$$R_1 = 330\Omega \quad R_3 = 4.7k\Omega$$

$$R_2 = 1k\Omega \quad R_4 = 2.2k\Omega$$

$$= 1498.55 + 248.12$$

$$= 1746.67\Omega$$

3.)

$$R_1 = 330 \begin{pmatrix} 1.05 \\ 0.95 \end{pmatrix} = 346.5 \leftarrow \text{max}$$

$$330 \begin{pmatrix} 1.05 \\ 0.95 \end{pmatrix} = 313.5 \leftarrow \text{min}$$

$$R_2 = 1000 \begin{pmatrix} 1.05 \\ 0.95 \end{pmatrix} = 1050 \leftarrow \text{max}$$

$$1000 \begin{pmatrix} 1.05 \\ 0.95 \end{pmatrix} = 950 \leftarrow \text{min}$$

$$\min R_1 \parallel R_2 = 235.41\Omega$$

$$\max R_1 \parallel R_2 = 260.53\Omega$$

$$④. R = \frac{V_s - V_{max}}{I_f}$$

$$R = \frac{3.3 - 2.0}{10 \text{ mA}} = 130 \Omega$$

$$⑤. a.) V_{out} = V_{REF} \left(1 + \frac{R_1}{R_2} \right)$$

$$V_{out} = 1.242 (1 + 0.59574) \approx 2.0 \text{ V}$$

$$b.) \text{ Max: } R_{1max} = 280 \text{ k} (1.01) = 282.8 \text{ k}$$

$$R_{2min} = 470 \text{ k} (0.99) = 465.3 \text{ k}$$

$$\text{Min: } R_{1min} = 280 \text{ k} (0.99) = 277.2 \text{ k}$$

$$R_{2max} = 470 \text{ k} (1.01) = 474.7 \text{ k}$$

$$V_{out \text{ max}} = 1.242 \left(1 + \frac{282.8}{465.3} \right)$$

$$V_{out \text{ max}} = 1.9856 \text{ V}$$

$$V_{in \text{ max}} = 1.235 \left(1 + \frac{277.2}{474.7} \right)$$

$$= 1.9562 \text{ V}$$

both round to approximately 2.0V

$$c.) T, 25^{\circ}\text{C} \rightarrow \pm 1\%$$

$$T, -40^{\circ}\text{C} \text{ to } 125^{\circ}\text{C} \rightarrow \pm 2\%$$

$$d.) \text{ efficiency} = \frac{I_o V_o}{(I_o + I_q) V_i} \times 100$$

$$= \frac{(0.5)(1.97)}{(0.5 + 0.008) \times 5} = \frac{0.9855}{2.54} = 38.8\%$$